

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Concept Mapping
Annexure A

Criteria	Concept Identification (2.5)	Linkages (2.5)	Organization (2)	Integration of Literature (2)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I _____ Siva Charan (192525039)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Siva Charan

RUBRICS FOR CALCULATION:

Concept Mapping					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Annexure A

Concept Map 1: Decision Support Systems (DSS)

Task Description:

Construct a concept map that explains how a Decision Support System (DSS) supports organizational decision-making.

□ What your map must show:

1. Core components: Data, Model, User Interface
 2. Types of DSS
 3. Flow: Data → Analysis → Decision
 4. Role in business decision-making
 5. Real-world example linkage
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Concept Map 2: Operational Systems vs DSS

Task Description:

Develop a comparative concept map that clearly distinguishes Operational Systems (OLTP) and Decision Support Systems (OLAP).

What your map must show:

1. Definitions of OLTP and DSS
2. Differences (data type, purpose, users, processing)
3. Similarities and interactions
4. Real-time vs historical data
5. Example systems (banking vs analytics)

Concept Map 3: Architecture of Data Warehouse

Task Description:

Create a hierarchical concept map illustrating the architecture of a Data Warehouse.

What your map must show:

1. Data Sources → ETL → Data Warehouse → Data Marts

2. Metadata layer
3. Front-end tools (reporting, OLAP)
4. Data flow direction
5. Role of each layer

Concept Map 4: OLAP, Data Cubes & Dimensional Modeling

Task Description:

Design a concept map that connects OLAP operations, Data Cubes, and Dimensional Modeling techniques.

☐ What your map must show:

1. Data cube structure (dimensions, measures)
2. OLAP operations (slice, dice, roll-up, drill-down)
3. Star schema and snowflake schema
4. Fact and dimension tables
5. Analytical usage

Concept Map 5: Query Processing & BI Tools (KNIME)

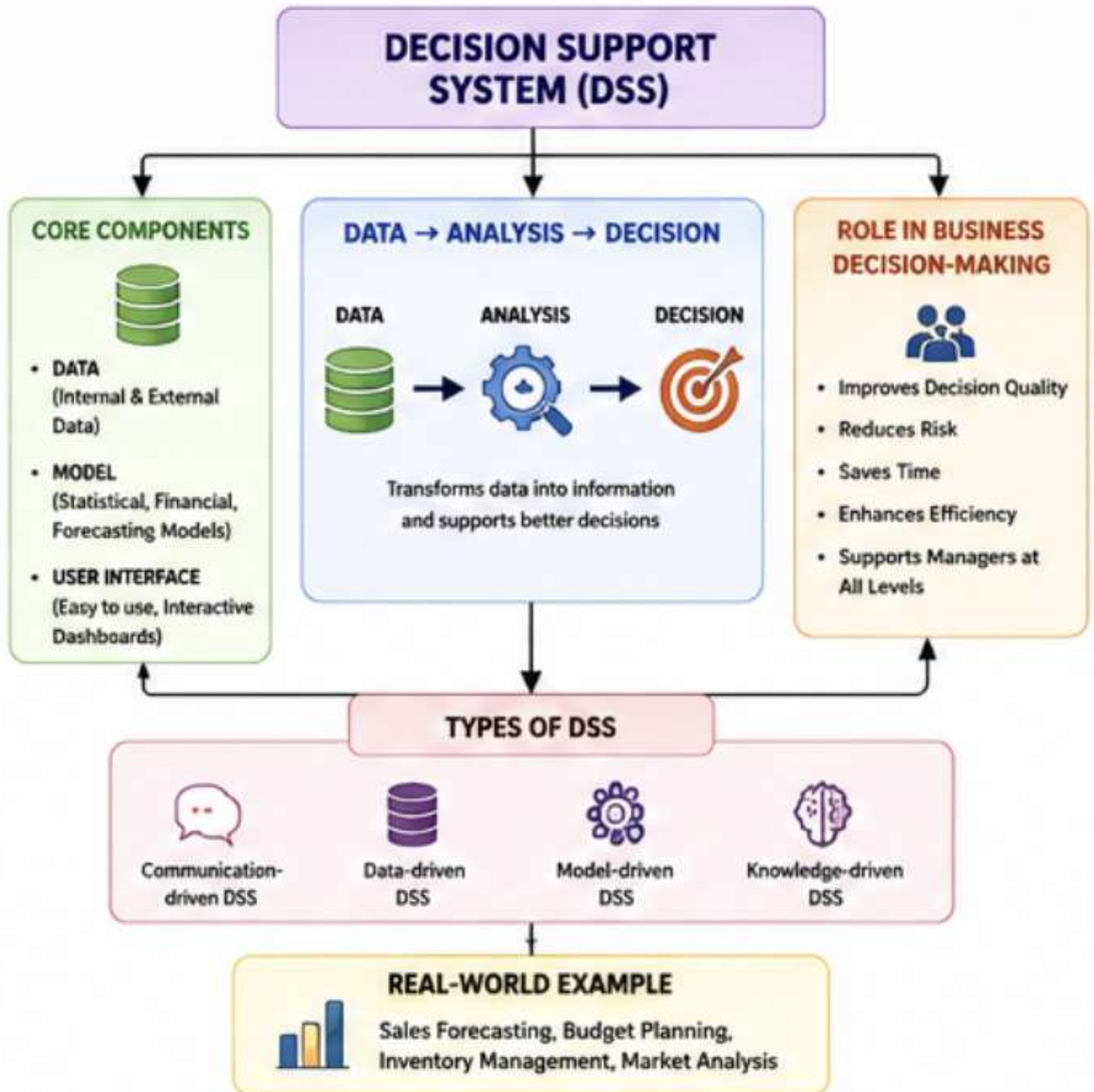
Task Description:

Construct a concept map explaining how query processing integrates with BI tools, focusing on KNIME for reporting and OLAP applications.

What your map must show:

1. Query processing stages (Parsing, Optimization, Execution)
2. Role of BI tools
3. KNIME workflow structure
4. Data → Query → Visualization pipeline
5. OLAP application integratio

1. CONCEPT MAP: DECISION SUPPORT SYSTEMS (DSS)



2. CONCEPT MAP: OPERATIONAL SYSTEMS (OLTP) vs DECISION SUPPORT SYSTEMS (OLAP)

OPERATIONAL SYSTEMS (OLTP)



Definition: Systems that handle day-to-day operations and transactions.

	Data Type	: Current, Detailed, Real-time Data
	Purpose	: Run the Business
	Users	: Clerks, Staff, Operational Users
	Processing	: High Volume, Simple Transactions
	Data Characteristics	: Up-to-date, Volatile
	Example Systems	: Banking Transactions, Order Processing, Payroll



DECISION SUPPORT SYSTEMS (OLAP)



Definition: Systems that analyze historical data to support decision-making.

	Data Type	: Historical, Summarized, Integrated Data
	Purpose	: Support Decision Making
	Users	: Managers, Analysts, Executives
	Processing	: Complex Queries, Aggregations
	Data Characteristics	: Historical, Subject Oriented
	Example Systems	: Sales Analysis, Market Analysis, Financial Reporting

SIMILARITIES



Both use data



Both support business



Both improve efficiency

INTERACTION

OLTP systems feed data to OLAP systems for analysis.

OLTP → Real-time Data

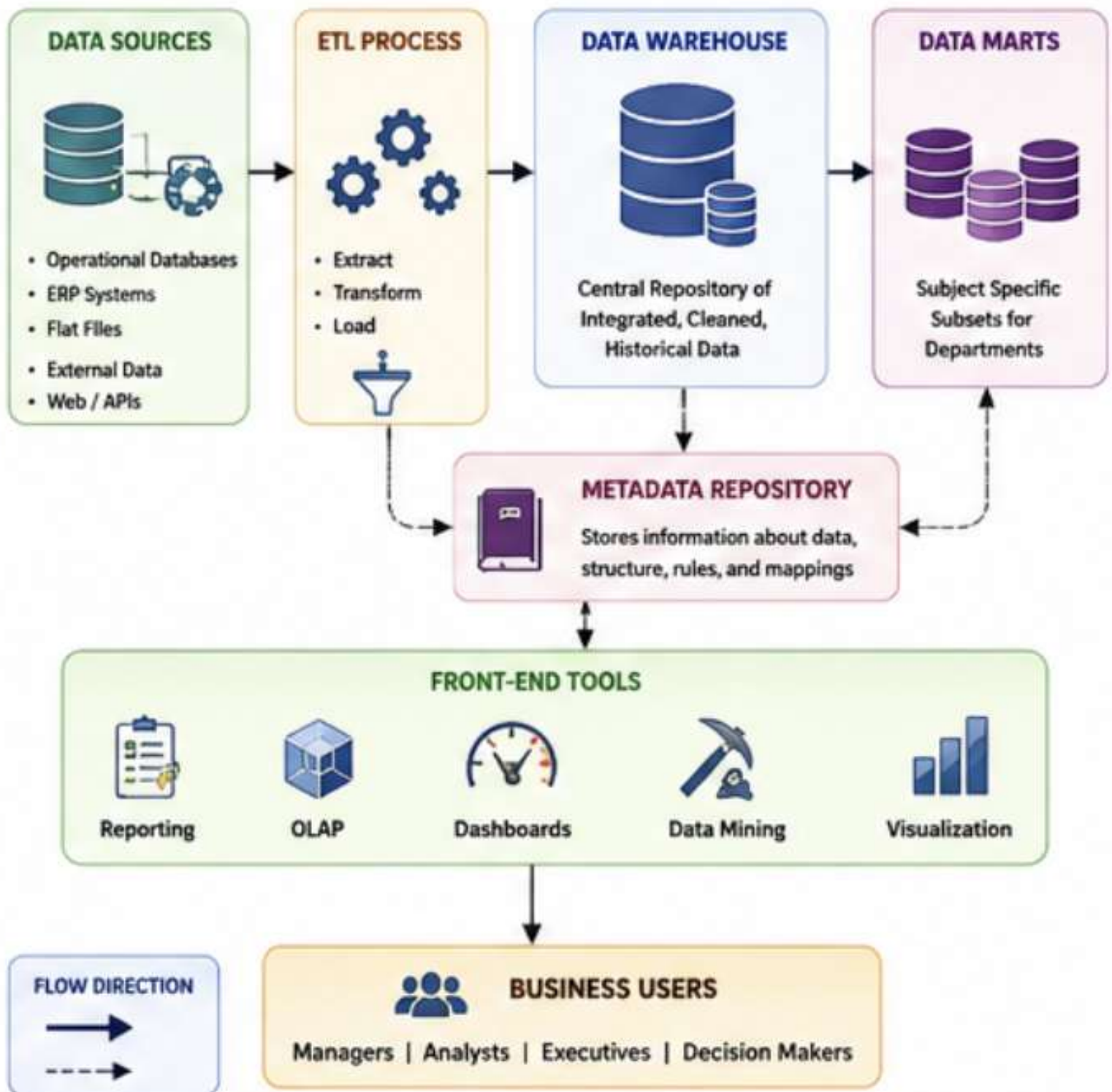
Used for immediate operations



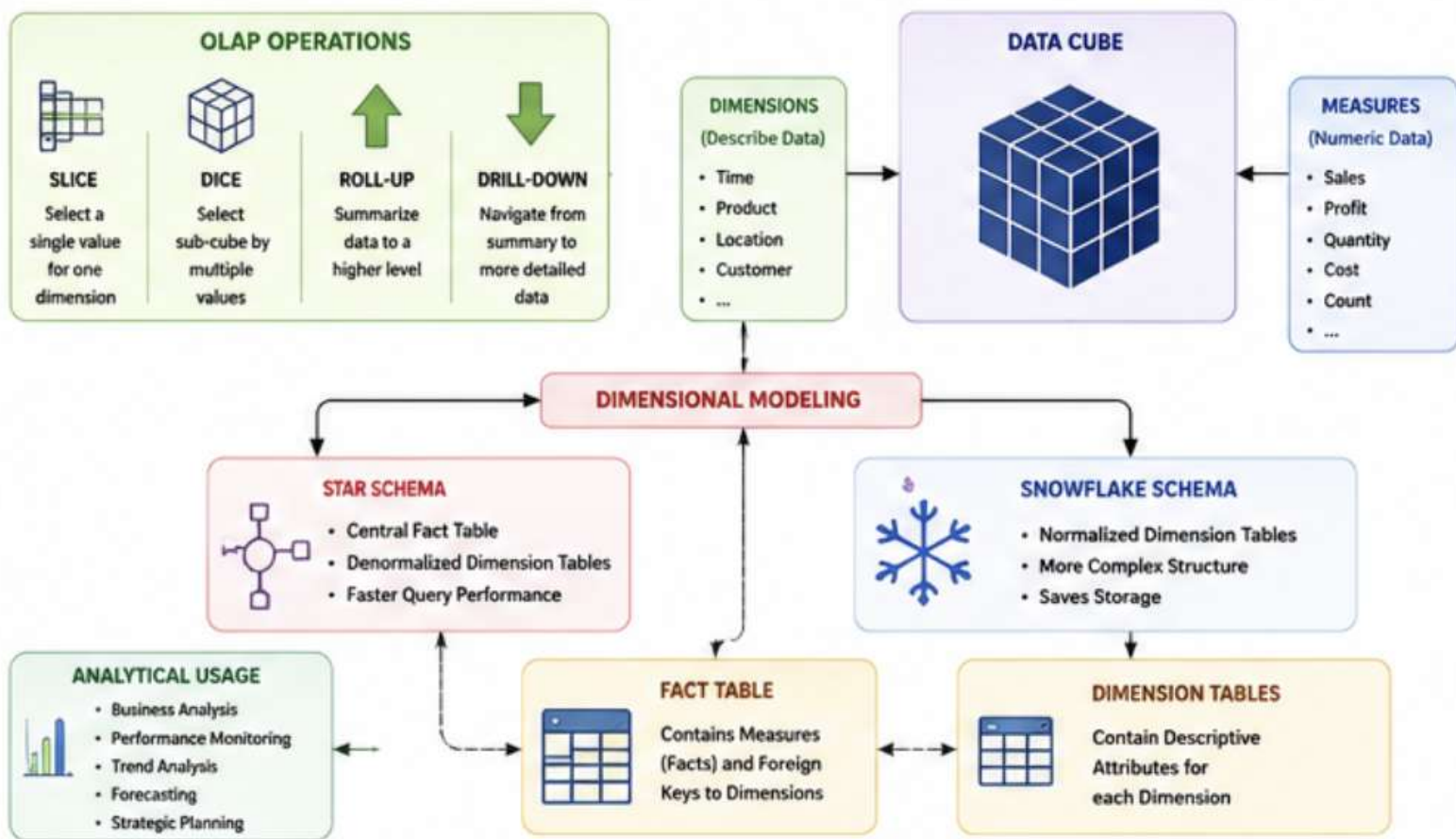
REAL-TIME vs HISTORICAL DATA

Used for analysis & planning

3. CONCEPT MAP: ARCHITECTURE OF DATA WAREHOUSE



4. CONCEPT MAP: OLAP, DATA CUBES & DIMENSIONAL MODELING



5. CONCEPT MAP: QUERY PROCESSING & BI TOOLS (KNIME)

